FORMULAS IN ACTION

A formula can be used to calculate how much interest (the amount a bank pays someone in exchange for being able to borrow their money) is paid into a bank account over a particular period of time. The formula for this is principal (or amount of money) × rate of interest × time. This formula is shown here.

this stands for principal, which just means the amount

this stands for rate of interest

$$I = PRT$$

this stands for the time it will take to earn interest

this stands for interest

There is a bank account with $500 in it, earning simple interest (see pp.74–75) at 2% a year. To find out how much time (T) it will take to earn interest of $50, the formula above is used. First, the formula must be rearranged to make T the subject. Then the real values can be put in to work out T.

▷ **Move P**
The first step is to divide each side of the formula by P to move it to the left of the equals sign.

$$I = PRT \Rightarrow \frac{I}{P} = RT$$

to remove P from the right side, divide each side of the formula by P

remember that dividing the right side by **P** gives **P**RT/**P**, but the **P**s cancel out, leaving **RT**

▷ **Move R**
The next step is to divide each side of the formula by R to move it to the left of the equals sign.

to remove R from the right side, divide each side of the formula by R

$$\frac{I}{P} = RT \Rightarrow \frac{I}{PR} = T$$

remember that dividing the right side by **R** gives **R**T/**R**, but the **R**s cancel out, leaving **T.**

▷ **Put in real values**
Put in the real values for I ($50), P ($500), and R (2%) to find the value of T (the time it will take to earn interest of $50).

interest (I) is $50

length of time (T) to earn interest of $50 is 5 years

$$T = \frac{I}{PR} \Rightarrow \frac{50}{500 \times 0.02} = \mathbf{5\ years}$$

principal (P) is $500

rate of interest (R) is 2%, written as a decimal as 0.02